

Introduction To Artificial Neural Network

Recognition: Detects emotions, faces, patterns, voice, and structures.

Creation: Generates realistic images, videos, and music.

The Microprocessor (Computer Chip): Runs your devices and does hard tasks. It is made of millions of tiny **logic gates** working together.

The Human Brain: Much more powerful than a computer chip. It is made of billions of living cells called **neurons**.

Artificial Brain: By copying how the brain connects cells and running it on a computer chip, we get an **Artificial Neural Network**.

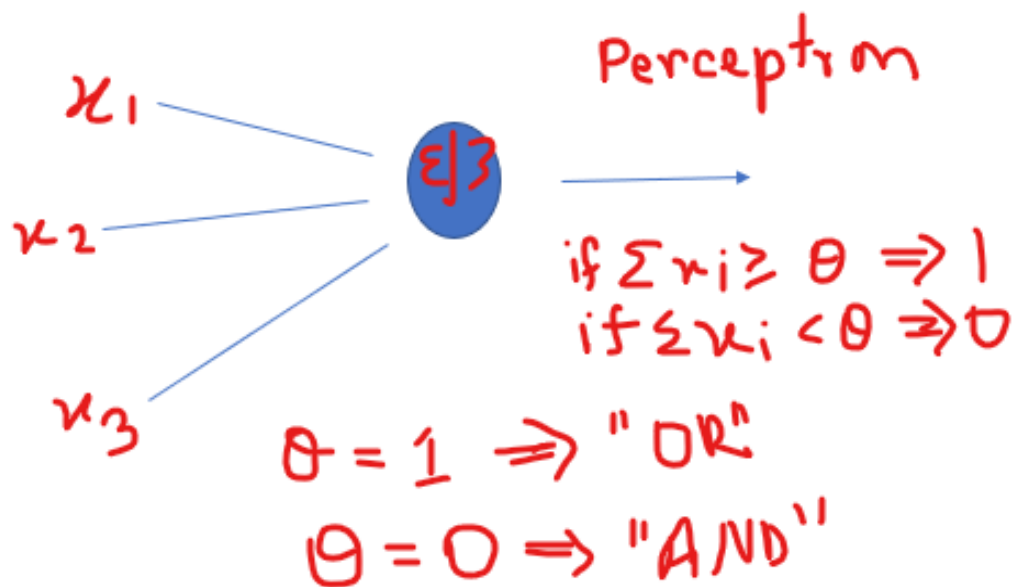


Perceptron Its Artificial Neuron

McCulloch-Pitts Neuron (1943): The very first mathematical model of a cell. It took inputs and gave a simple yes/no answer, but it could not learn by itself.

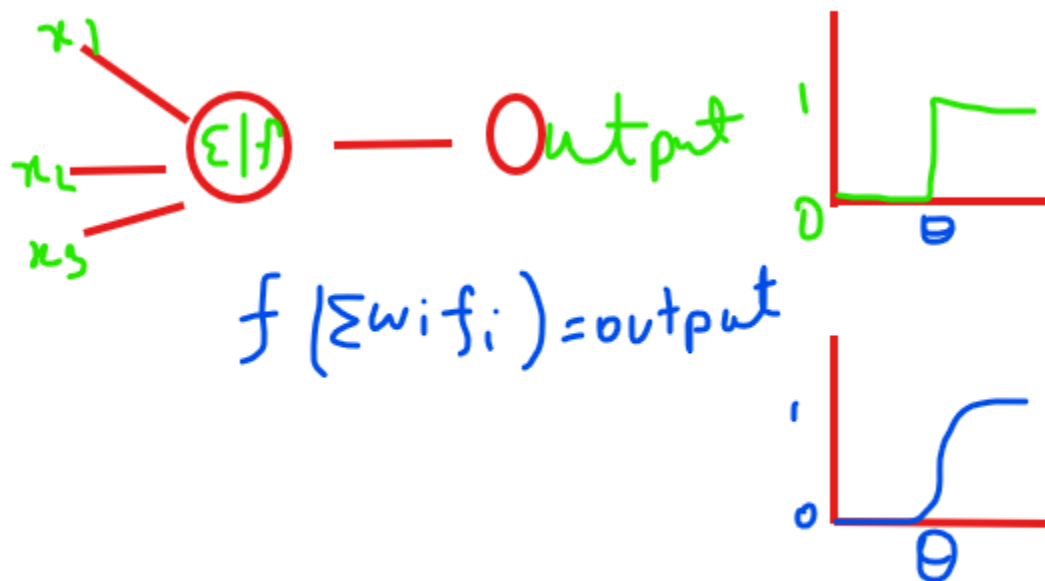
Rosenblatt's Perceptron (1957): The first real artificial neuron that could learn. It added **Weights** (giving different importance to different inputs) to fix its own mistakes.

Bias: A special number added to the neuron to help it adjust its decision-making threshold smoothly.

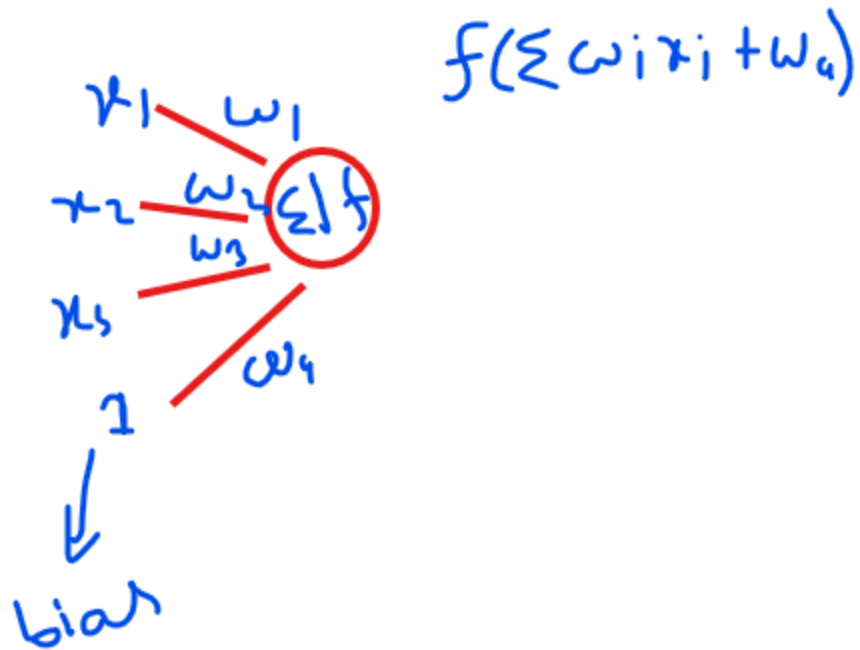


Perceptron

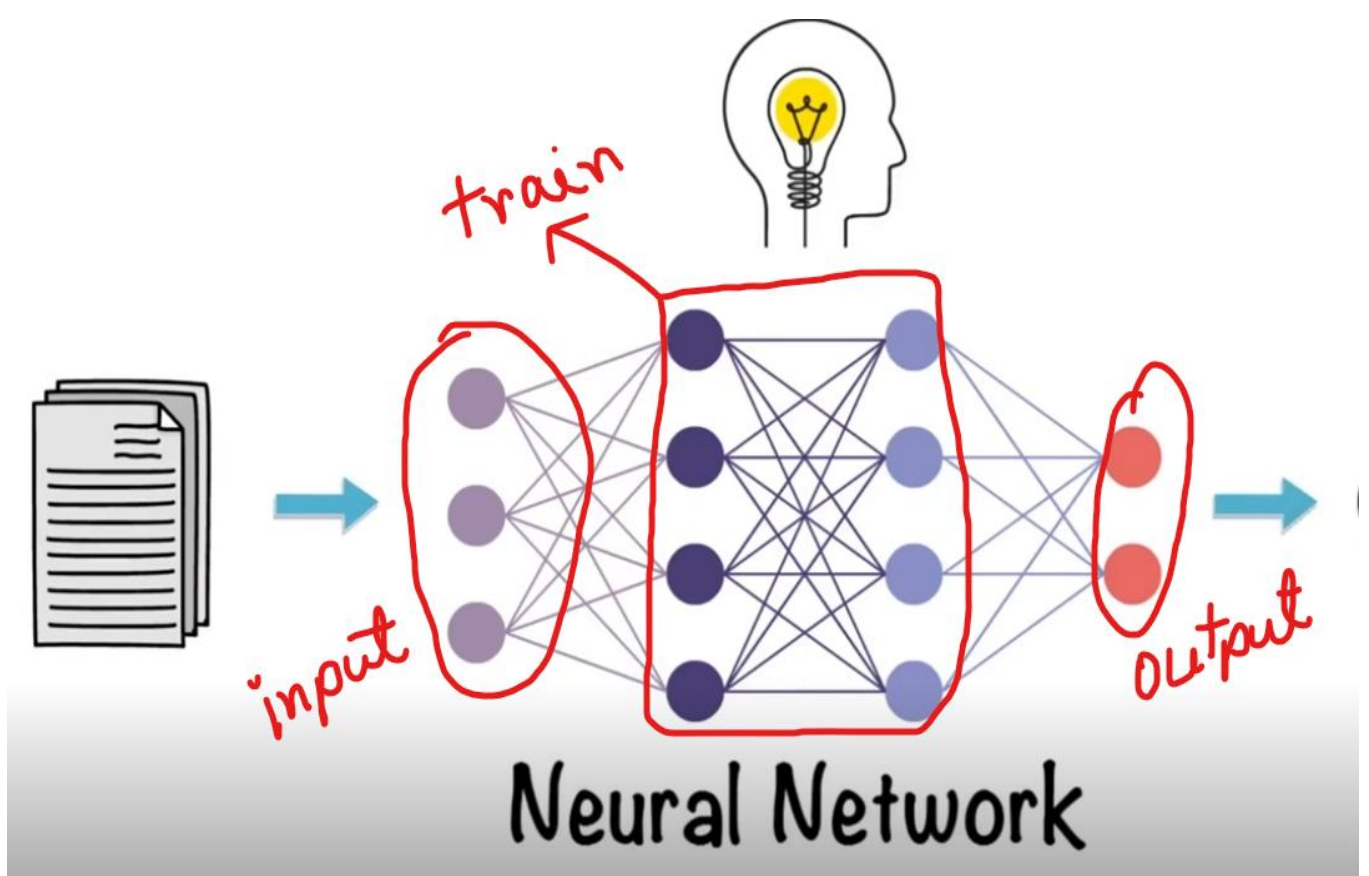
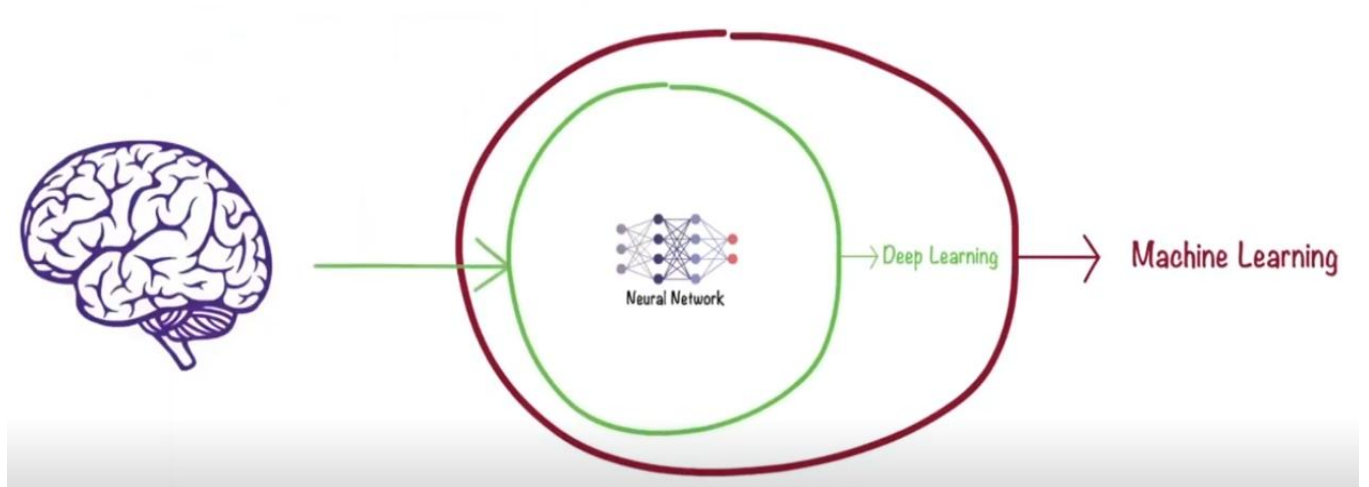
⇒ What Does It Do?

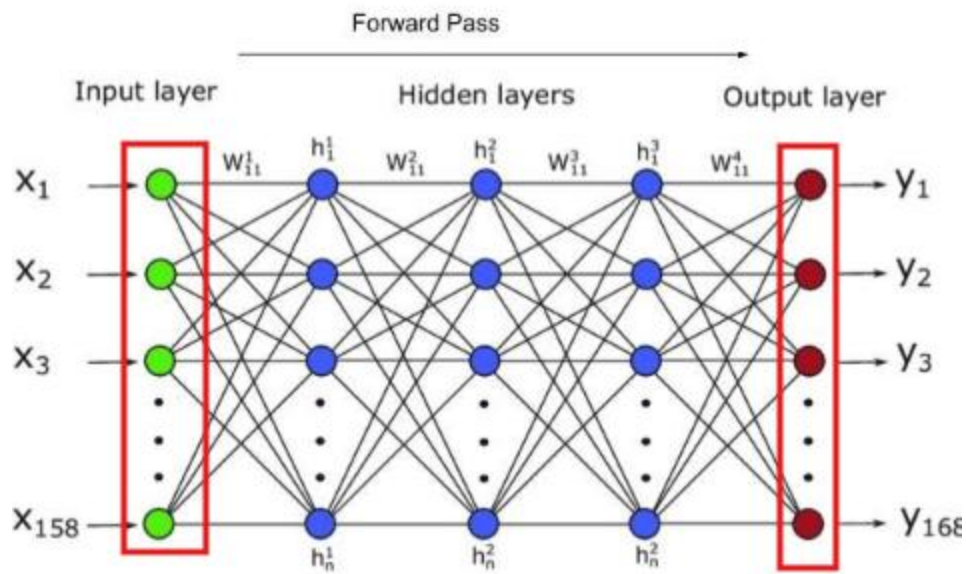


⇒ Bias

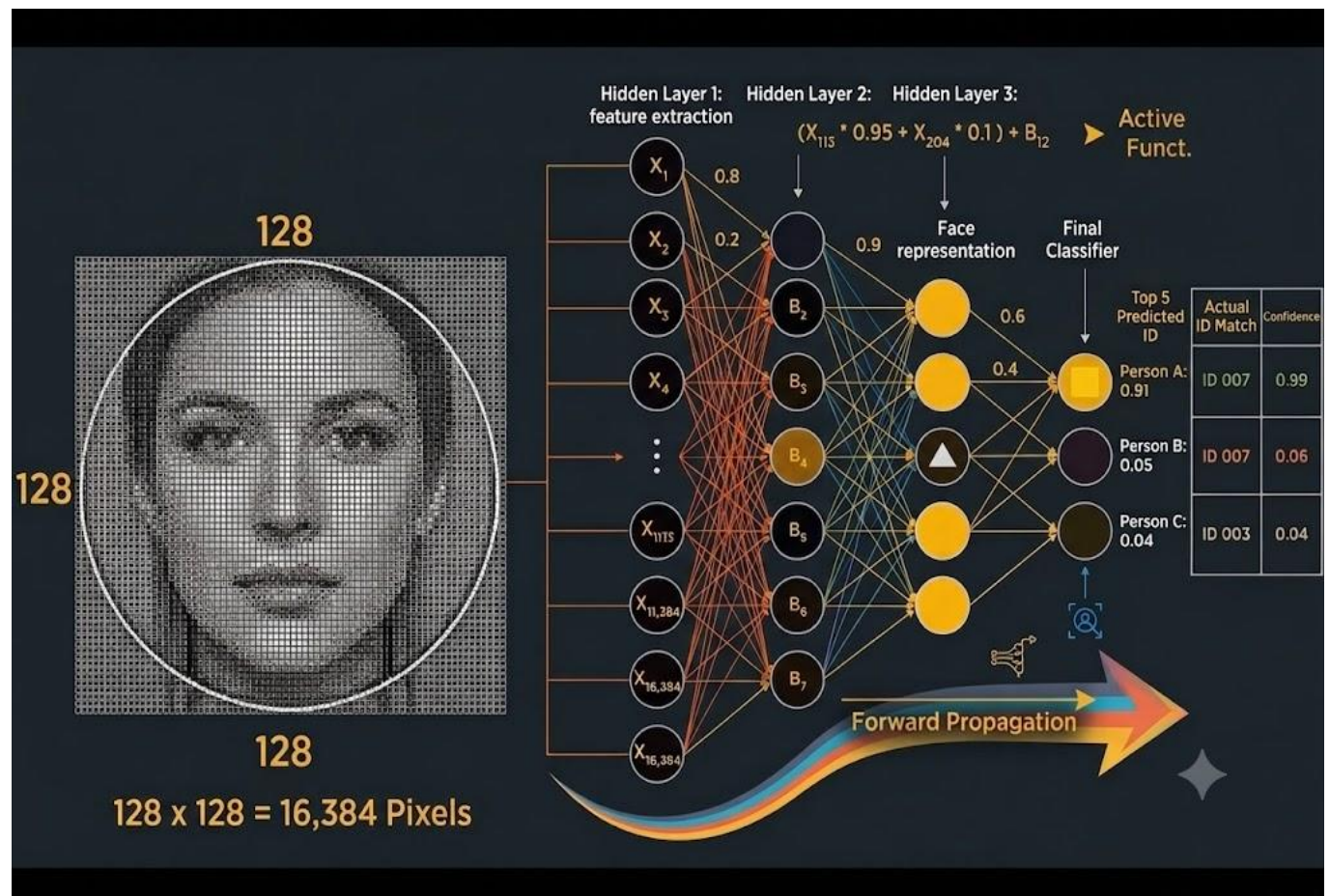


⇒ "The Embryo Of An Electronic Computer That Expects Will Be Able To Walk , Talk, See, Write, Reproduce Itself And Be Conscious Of Its Existence". -Rosenblatt, 1958

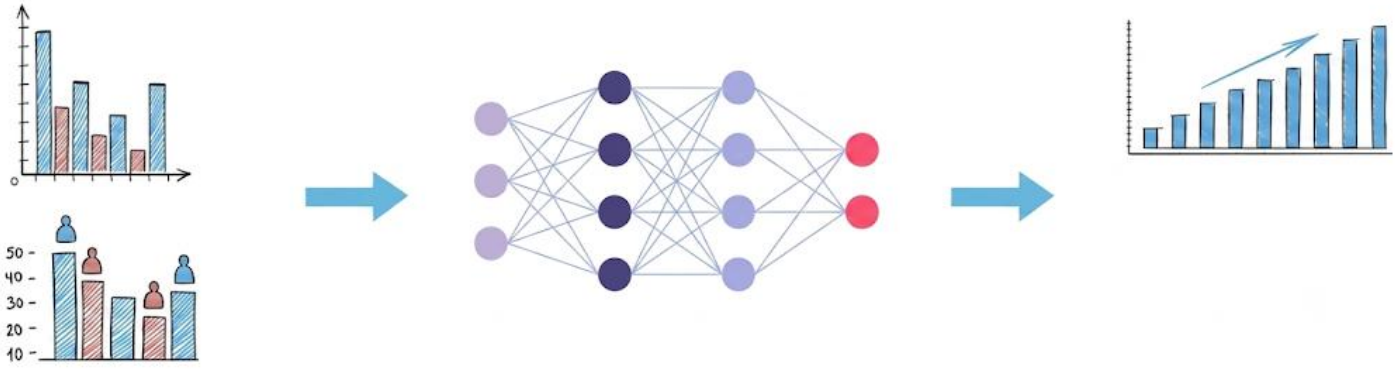




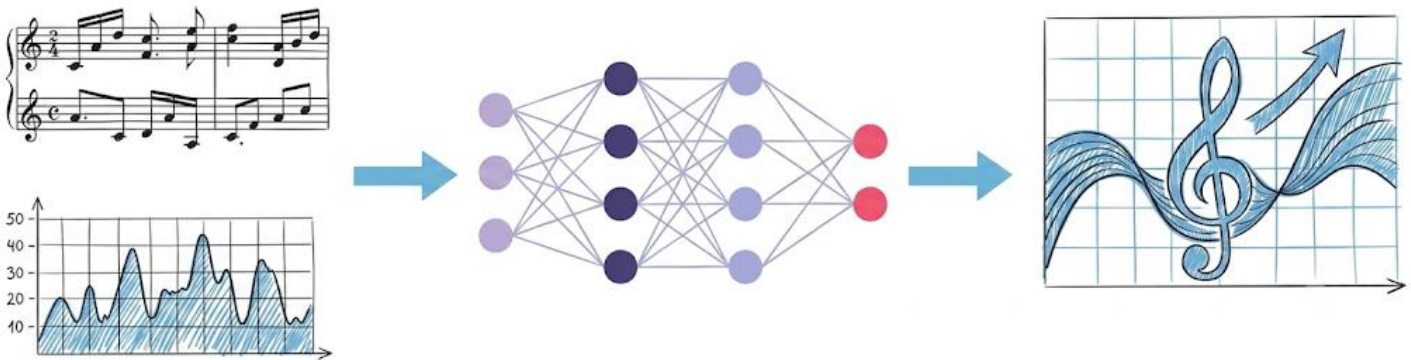
FACIAL RECOGNITION



FORECASTING



MUSIC Composition



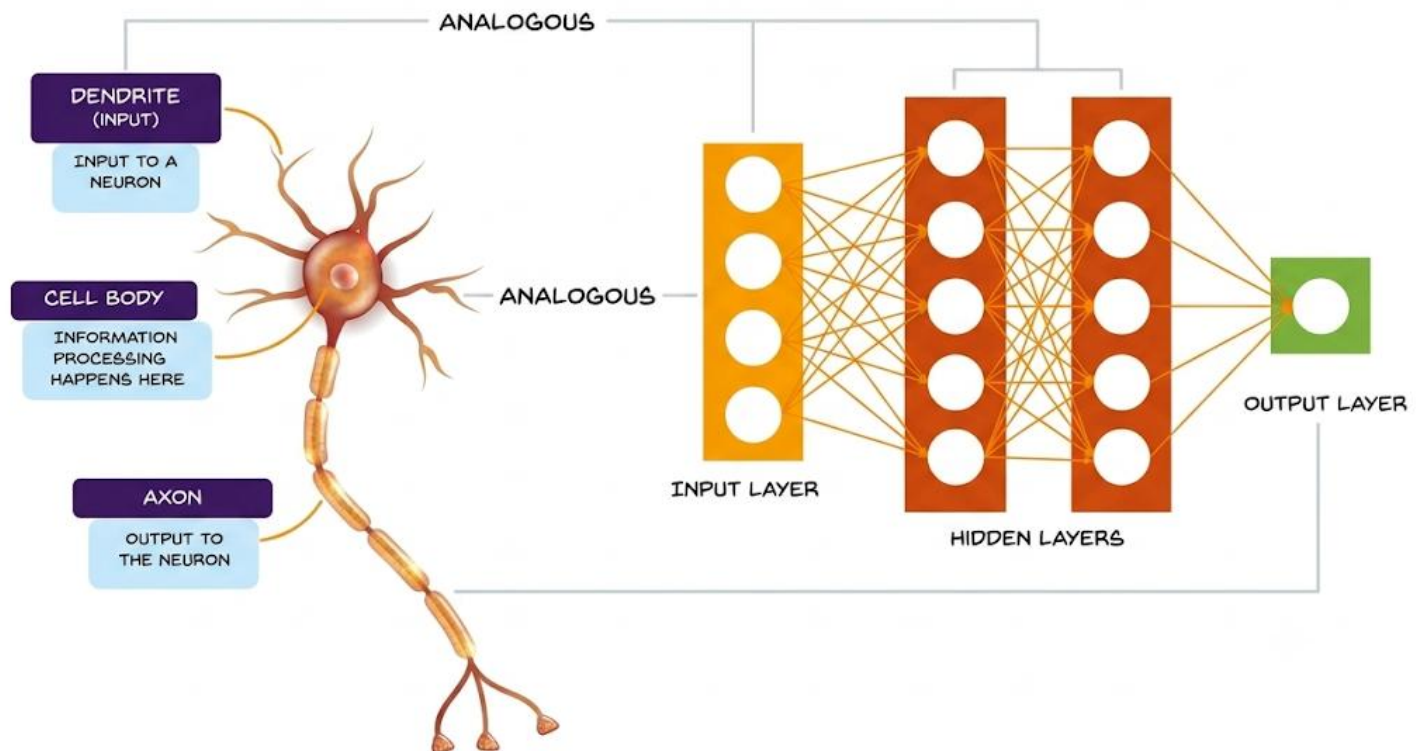
What Is DEEP Learning?

→ DEEP Learning Is A Machine Learning Technique That Teaches Computers To Do What Comes Naturally To Humans Learn By Example.

What Is Artificial Neural Network?

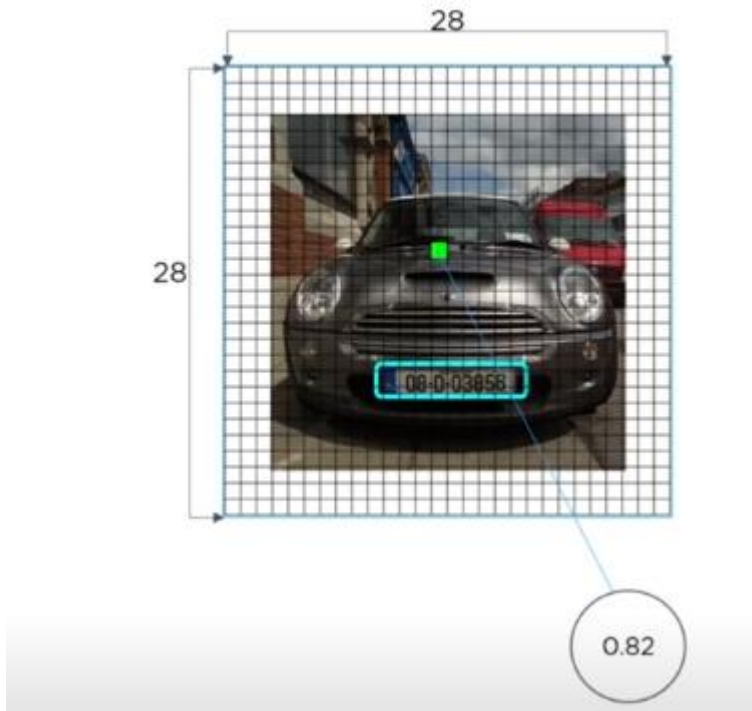
→ A Neural Network Is A System Of Hardware AND/OR Software Patterned After The Operation Of New Year Neurons In The Human Brain Neural Network Also Called Artificial In Neural Networks Is A Way Of Achieving Deep Learning.

COMPARISON OF BIOLOGICAL AND ARTIFICIAL NEURONS



How does a Neural Network Work?

- **Input layer** :- Picks up input signals and passes them to the next layer
- **Hidden layer** :- Does all kinds of calculations and feature extraction
- **Output Layer**:- This layer delivers the final result



This picture has 28×28 pixel its input signal and in the hidden layer it train the function and delivered the output.

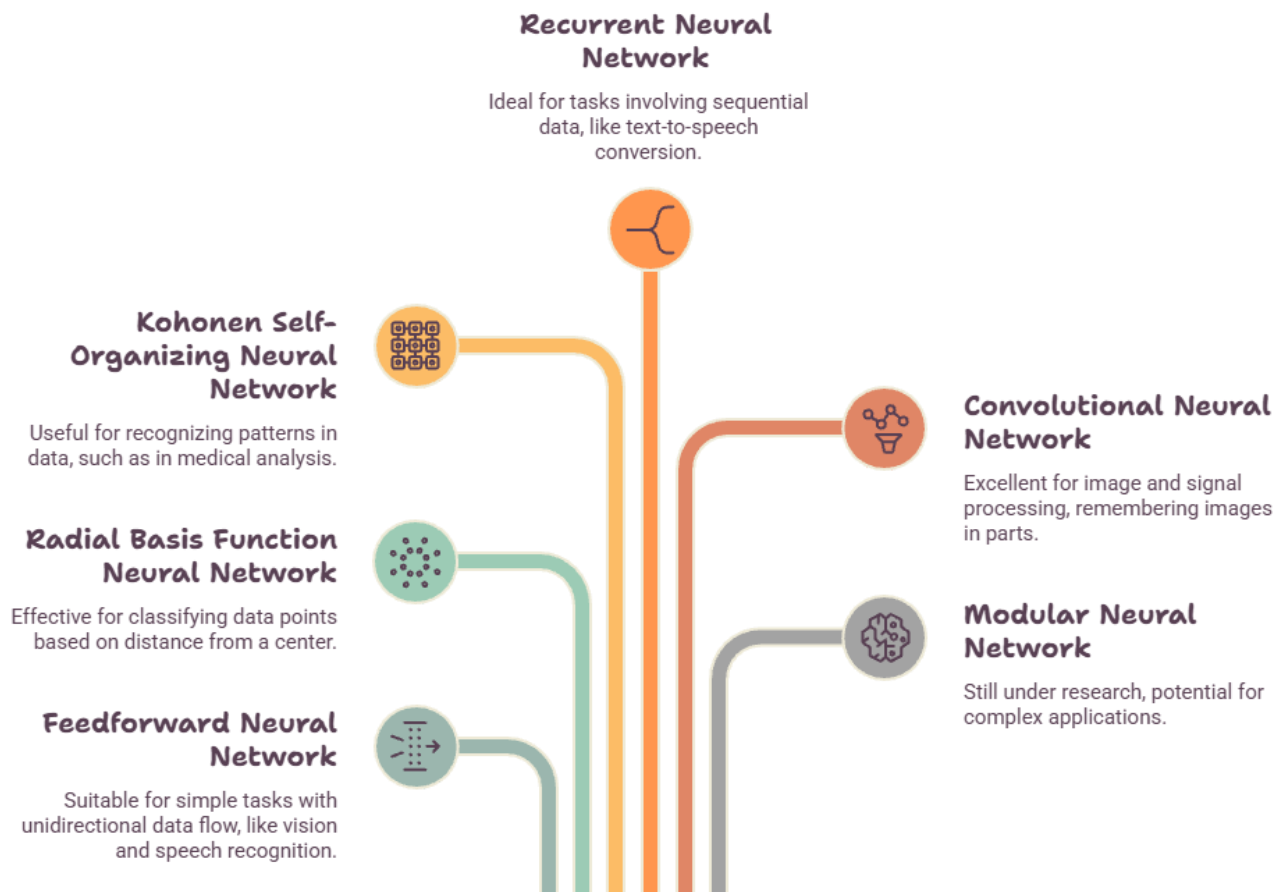
Traffic camera work like that analysis the number plate.

Typers Of Artificial Neural Network ?

- Feedforward neural network
 - ✓ simplest form of an data travels only in one direction
 - ✓ Applications vision and speech recognition
- Radial basis function neural network
 - ✓ This model classifies the data point based on its distance from a center point
- Kohonen self organizing neural network
 - ✓ Vectors of random dimensions are input to discrete map comprised of neurons
 - ✓ Application used to recognize patterns in data like in medical analysis
- Recurrent neural network
 - ✓ This hidden layer saves its output to be used for future prediction
 - ✓ Application text to speech conversion model
- Convolution neural network

- ✓ The input features are taken in batches like a filter. This allows the network to remember an image in parts
- ✓ Application used in signal and imaged process
- Modular neural network
 - ✓ Application still undergoing research

Which type of neural network should be used for a specific application?



Made with Napkin

ADVANTAGES OF ARTIFICIAL NEURAL NETWORK

- ✚ It's a limited entally by inputs and result given to them initially by an expert system.
- ✚ This everything in handy for robotics and pattern recognition system.
- ✚ Artificial neural networks have the potential for high fault tolerance.
- ✚ Artificial neural network networks are capable of debugging or diagnosing a network on their own.

- ✚ Nonlinear system have the capability of finding shortcuts to reach computationally expensive solution.

Applications of Artificial Neural Networks (ANN)

Handwriting Recognition

- ✚ Used to convert handwritten characters into digital characters.
- ✚ Allows the system to easily recognize and process manual text.

Stock Exchange Prediction

- ✚ Examines a massive volume of variables and market factors simultaneously.
- ✚ Predicts stock prices on a daily basis, serving as a powerful tool to help stockbrokers make decisions.

Travelling Salesman Problem

- ✚ Assists drivers and logistics systems in finding the optimal path to travel between all cities in an area.
- ✚ Helps solve this complex problem by providing higher efficiency at a minimal cost.

Image Compression

- ✚ Used as a data compression tool.
- ✚ Successfully stores and recreates the actual image again without losing essential data.

What is a CNN?

A **Convolutional Neural Network (CNN)** is an AI system built to analyze images. Instead of looking at a whole picture at once, it breaks it down piece-by-piece—detecting lines, edges, and shapes—until it can identify the object.

How Computers See Images

To a computer, an image is just a grid of numbers called **pixels**:

- **Black & White:** A single 2D grid of numbers.

- **Color (RGB):** A 3D stack of three grids (Red, Green, and Blue).

Convolution Layer (Feature Detector)

- A small window called a **filter** slides across the image pixel by pixel.
- It calculates math at each stop to find specific patterns like edges or curves, creating a map of where those features are located.

ReLU Layer (The Cleaner)

- It looks at the feature map and changes all negative numbers to zero, while keeping the positive numbers.
- It removes unnecessary "noise" and highlights only the areas where a feature was successfully found.

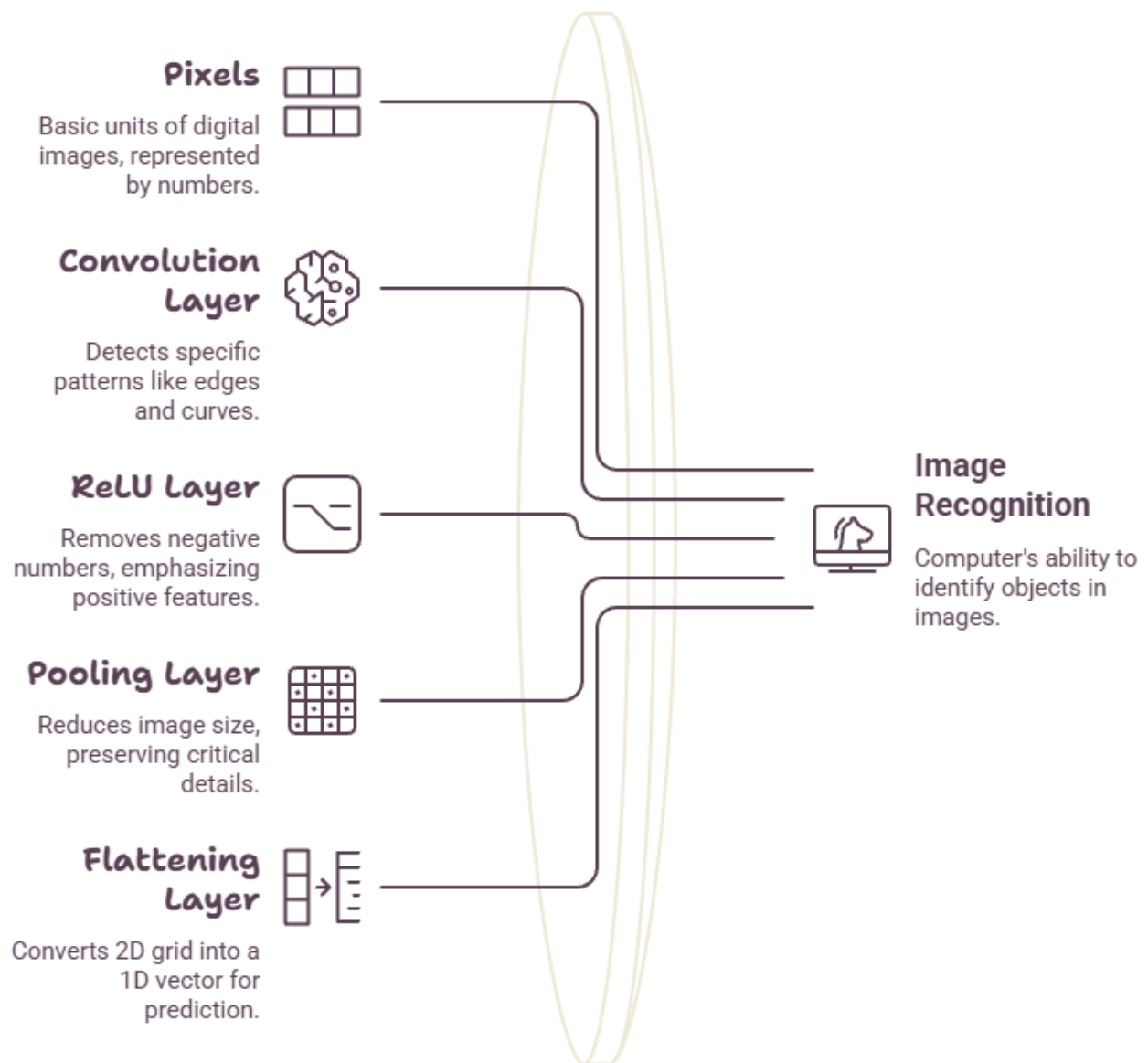
Pooling Layer (The Shrinker)

- It reduces the size of the grid. **Max Pooling** slides a small box over the data and keeps only the highest number, throwing the rest away.
- It shrinks the data to save computing power while keeping the most critical details.

Flattening Layer (The Unroller)

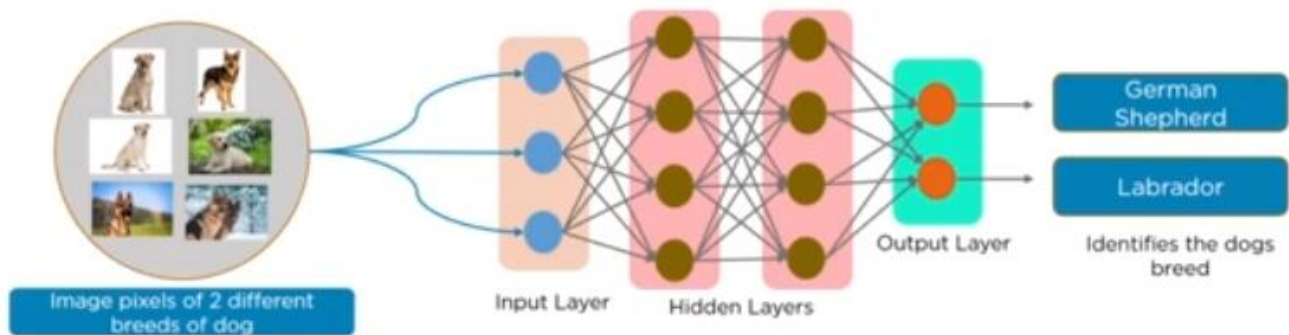
- It takes the final compressed 2D grid and unrolls it into a single, straight line of numbers (a 1D vector).
- This long line of numbers is sent to the final traditional network layer to make the ultimate prediction (e.g., "98% chance this is a cat").

Building Blocks of Computer Vision





Neural Networks used in Deep Learning, consists of different layers connected to each other and work on the structure and functions of a human brain. It learns from huge volumes of data and uses complex algorithms to train a neural net.



POPULAR NEURAL NETWORK

Neural Network Types

Recurrent Neural Network

Used for speech recognition.



Feedforward Network

Used in general regression and classification problems.



Deep Belief Network

Used for cancer detection.



Convolutional Network

Used for image recognition tasks.



Deep Neural Network

Used for acoustic modeling applications.



Speech Recognition

- Converts spoken audio into written text.
- Voice assistants like Siri, Google Assistant, or Alexa understanding your voice commands.

Machine Translation

- Translates text from one language to another.
- Google Translate predicting the next word in a sentence to provide accurate, grammatically correct translations.

Sentiment Analysis

- Analyzes a block of text to determine the underlying emotion (positive, negative, or neutral).
- Tracking customer reviews or social media comments to see if people like a product.

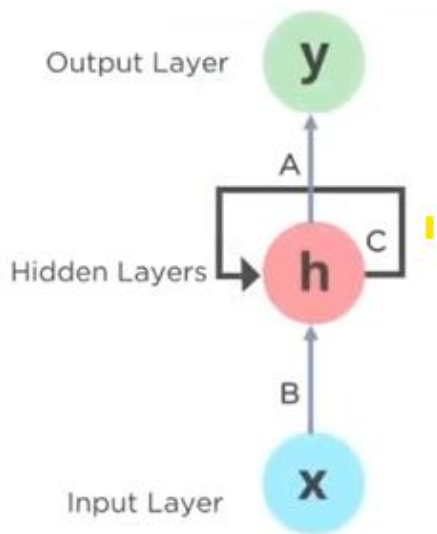
Time Series & Stock Prediction

- Analyzes historical, chronological data to predict future trends.
- Predicting stock market prices, weather forecasting, or anticipating electricity grid demands.

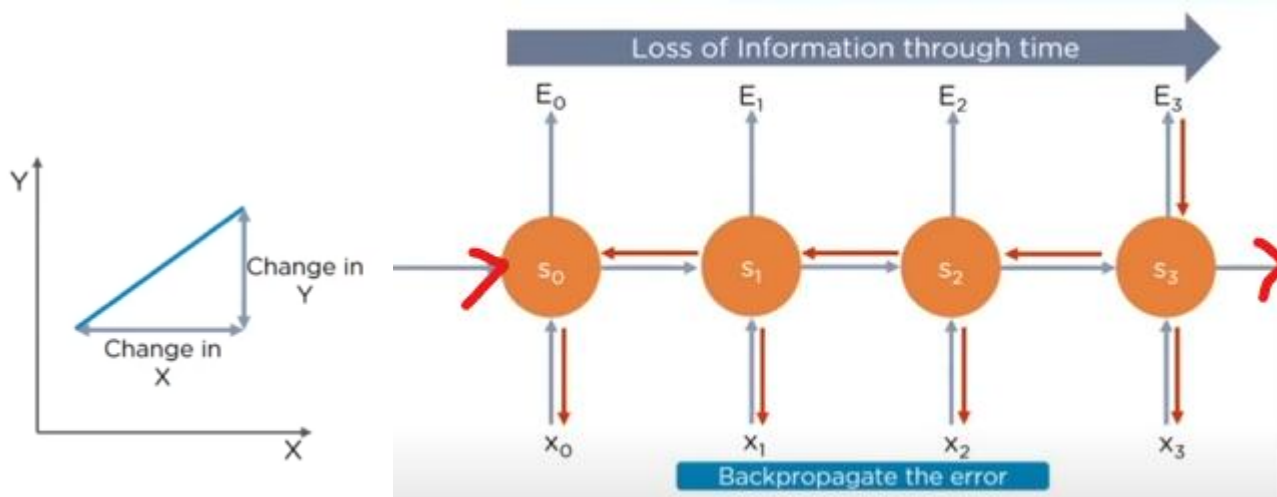
Text Generation & Autocomplete

- Predicts the most likely next word or character based on what you have already typed.
- Smart compose in Gmail or predictive text on your smartphone keyboard.

RNN look like

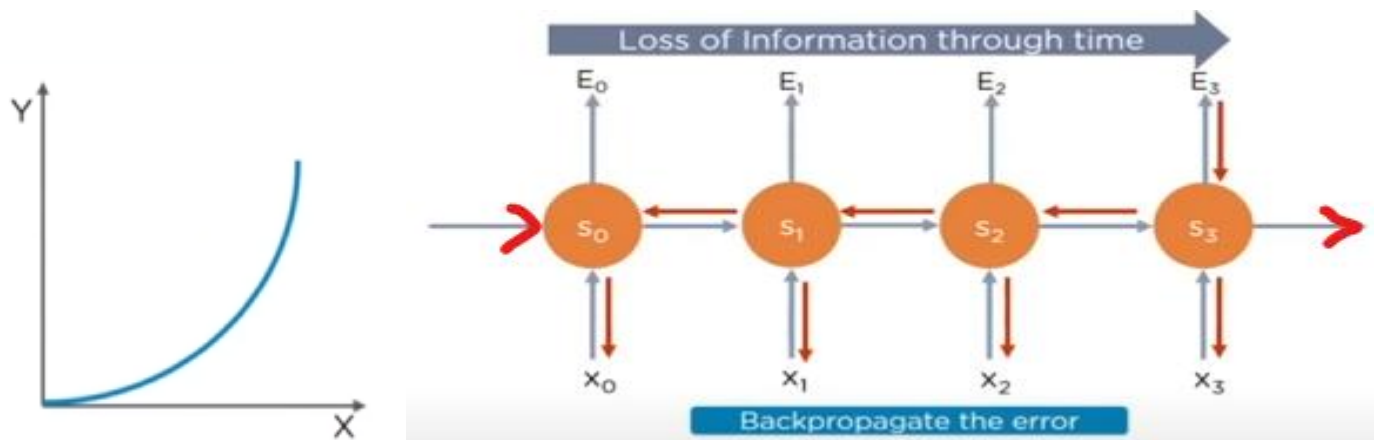


VANISHING GRADIENT PROBLEM



While training a RNN, your slope can be either too small or very large and this makes training difficult. When the slope is too small, the problem is known as *Vanishing gradient*.

EXPLODING GRADIENT PROBLEM



When the slope tends to grow exponentially instead of decaying, this problem is called *Exploding gradient*.

SOLUTION TO GRADIENT PROBLEM



LSTM (Long Short-Term Memory) is a smart type of AI network that has an excellent long-term memory.

Traditional networks are forgetful and struggle with long sequences of data. LSTMs solve this by using three internal "gates" to manage their memory like a conveyor belt.